Forecasting Café Ingredient Usage from Heterogeneous Databases with Hand-Built Exponential Smoothing

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***Abstract*—Small businesses such as cafés need to know how much of each ingredient they will consume in the coming days in order to plan purchases and avoid both stock-outs and spoilage. We demonstrate a lightweight forecasting tool that predicts the *daily usage of individual ingredients* such as milk, sugar, or coffee grounds by joining transactional sales facts stored in PostgreSQL with drink recipes stored in MongoDB, and by forecasting the resulting time series with an exponential-smoothing method (Single and Triple Exponential Smoothing / Holt-Winters) that we implemented from scratch. Users can switch ingredients, retune the smoothing parameters, and immediately see both a future forecast and a validated backtest with error metrics. The demo shows how classic, explainable, non-ML forecasting combined with polyglot persistence already yields actionable planning support.**

***Index Terms*—forecasting, time series, exponential smoothing, Holt-Winters, polyglot persistence, demand planning**

I. Introduction

Food or drink retailers face a recurring planning problem: how much of each ingredient to buy for the next few days. Order too little and popular drinks sell out; order too much and milk spoils. In a café the data that are actually recorded are *sales of finished drinks*. Projecting it into the future, is a small but representative instance of forecasting demand, a task that appears in supply chains, energy load planning, and many other areas far beyond coffee [4]. The coffee shop here simply acts as a concrete, relatable example.

We tackle the challenge with a transparent, classic forecasting pipeline instead of a black-box model. Daily drink sales are aggregated in the database, expanded into per-ingredient demand through the recipes, and extrapolated with self-implemented exponential smoothing. Because the method is simple and explainable, every prediction can be validated against held-out history, and the café owner can understand *why* the tool recommends a given amount.

II. System Overview

Fig. 1 shows the architecture. The data live in two complementary databases, reflecting their different shapes—an instance of polyglot persistence [5]. **PostgreSQL** stores the transactional sales facts (one row per sale) and is well suited for the `GROUP BY` aggregation that produces a daily count per drink. **MongoDB** stores the drink recipes as documents, whose naturally nested ingredient lists are extracted with an aggregation pipeline. Both are needed: sales tell us *how many* of each drink were sold per day, while recipes tell us *how much* of an ingredient each drink consumes.

An external Python program orchestrates the pipeline. For a chosen ingredient it computes, for every day d,

$$\mathrm{usage}(d) = \sum_{\text{drinks } c} \mathrm{sales}(d, c) \cdot \mathrm{amount}(c), \tag{1}$$

where $\mathrm{sales}(d, c)$ comes from PostgreSQL and $\mathrm{amount}(c)$ from the recipe (0 if the drink does not use the ingredient).

The series is then forecast with additive exponential smoothing, introduced by Holt [3] for level and trend and extended to seasonality by Winters [2]. We implement two variants ourselves. Single Exponential Smoothing (SES) tracks only the level and predicts a flat line. Triple Exponential Smoothing (Holt-Winters) additionally smooths a trend and a per-weekday seasonal component,

$$\ell_t = \alpha\,(x_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}), \tag{2}$$
$$b_t = \beta\,(\ell_t - \ell_{t-1}) + (1-\beta)\,b_{t-1}, \tag{3}$$
$$s_t = \gamma\,(x_t - \ell_t) + (1-\gamma)\,s_{t-m}, \tag{4}$$

with season length $m = 7$, updating the seasonal index against the current level ℓ_t as in Winters' original method [2]. We use the additive form because the series contains zero-demand days, for which the multiplicative variant is numerically undefined. Our café series shows a clear weekly season (weekends differ from weekdays, visible in Fig. 2) and a mild trend—the situation in which SES is known to underperform [1] so Holt-Winters is our default and SES serves as a baseline.

III. Demonstration Workflow

The forecasting tool is accessed from a small command-line interface: users pick one or more ingredients, choose SES or

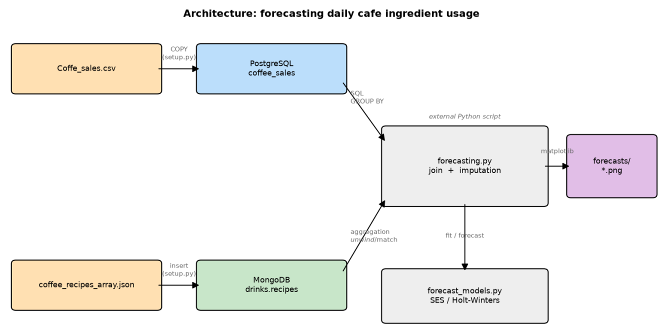

Fig. 1. Architecture: PostgreSQL (sales facts) and MongoDB (recipes) feed an external Python script that joins, imputes, forecasts, and plots.

Holt-Winters, and adjust the smoothing factors, watching how the forecast and its error change.

A. *Setup*

The demo runs on a laptop under Ubuntu 24.04 (Intel Core i7-1165G7, 4 cores, 16 GB RAM). The two databases (PostgreSQL 18 and MongoDB 8) are provisioned via Docker Compose and preloaded with roughly one year (388 days) of café sales and the corresponding recipes. The forecasting CLI is a pure Python 3 script (`pymongo`, `psycopg`, `matplotlib`); no forecasting library is used, all model code is our own. Forecasts complete in seconds on this hardware, so the interaction is immediate.

B. *Workflow*

The command line interface is accessed using `python forecasting.py --help` which gives an overview of available ingredients to be forecasted as well as the parameters `--alpha/--beta/--gamma` that can be adjusted. As an example, running `python forecasting.py --milk`, trains on the whole history and projects the next 30 days using the Holt-Winters algorithm as its default forecasting method. As can be observed in the resulting plot (Fig. 2), the forecasted usage is visibly oscillating, similarly to the sales data. Besides the future projection, an *evaluation* plot is being generated, which holds out the last four weeks, forecasts them, and overlays the prediction on the true values together with RMSE, MAE, and MAPE (summarised across ingredients in Table I), so the user can see how close the model got. Another option for the default Holt-Winters algorithm, is the Simple Exponential smoothing which can be choosen using `--model ses` which disregards seasonality entirely. A takeaway is that no single model always wins (Table I): Holt-Winters beats SES on the strongly seasonal milk and sugar series, whereas the flatter, noisier coffee-grounds series is fit marginally better by plain SES. Percentage errors can look large on low-volume ingredients (e.g. MAPE $\approx$ 71% for sugar) simply because many days carry near-zero demand, which is why RMSE and MAE are being reported alongside.

TABLE I
BACKTEST ERROR ON A 28-DAY HOLD-OUT WITH DEFAULT PARAMETERS (α=0.3, β=0.05, γ=0.3, m=7). LOWER IS BETTER; BEST PER METRIC IN BOLD.

Ingredient	HW RMSE	SES RMSE	HW MAE	SES MAE
Milk	**669**	888	**513**	702
Sugar	**15.6**	17.1	**12.7**	13.9
Coffee grounds	5.2	**4.1**	4.4	**3.9**

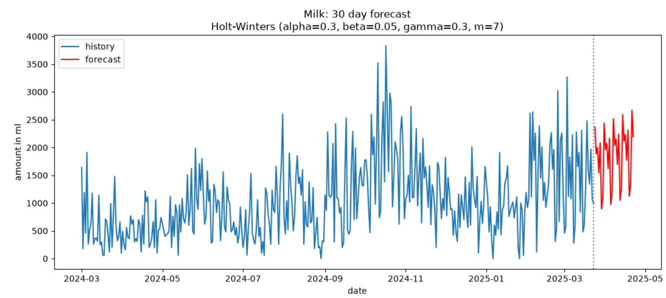

Fig. 2. Future forecast of daily milk usage (Holt-Winters), showing the projected weekly pattern beyond the last observed day.

C. *Conclusion*

Our demo shows that combining polyglot persistence with a small, self-built, fully explainable exponential-smoothing forecaster already gives a café actionable, per-ingredient purchasing guidance. Its unique selling point is transparency paired with interactivity: every prediction is validated against real history, and visitors can change the model and its parameters and see the effect on accuracy. The tool currently produces point forecasts with user-set rather than automatically optimised smoothing parameters; adding parameter search and prediction intervals would be suitable extensions.

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